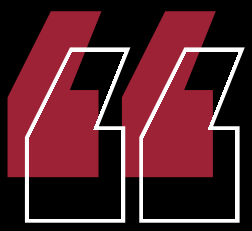


PROFESSIONAL EDUCATION

Digital Plus+ Programs

Digital Transformation:

From AI and IoT to Cloud, Blockchain, and Cybersecurity



Five technologies are redefining both the ways in which we update our products and the types of opportunities that exist in the market

Dr. Abel Sanchez

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Technology Today

The digital economy is immersed in a rapid evolution that affects every aspect of our lives, and this is only the beginning of an accelerating advancement.

The situation has carried professionals to a position with a lack of answers, which creates difficulties in maintaining updates and leading change in organizations

To remain at the forefront of this change, the leaders of the future must focus their projects on two fundamental pillars:

1. **Staying ahead of the competition**
2. **Taking advantage of new technologies**



Why Elevate Your Digital Strategy?

64%

In 2019, the businesses with the most digital development were 64% more likely to achieve their main commercial objectives of 2018.

Source: Adobe & Econsultancy

45%

of organizations reported a positive business impact and a greater profits due to digital transformation efforts.

Source: Deloitte

76%

of executives agree that organizations need to drastically redesign the experiences that unite technology and people in more human-centered ways.

Source: Accenture

The Relevance of The Digital Transformation

Blockchain, the cloud, artificial intelligence, the Internet of Things, and cybersecurity. These are the five vanguard technologies that are revolutionizing the way business works and changing key aspects of our society. Understanding and utilizing these technologies allows professionals from all sectors to not only guide their organizations through these changes, but to use them to their advantage.



About this Course



The course Digital Transformation: From AI and IoT to Cloud, Blockchain, and Cybersecurity analyzes the history of these innovative technologies and prepares participants to successfully and practically take on the digital transformation.

Participants will become familiar with the technologies that are currently opening the doors to the digital transformation, guided by two MIT professors and leaders in the field.

1 **Understanding technologies**

2 **Implementing them in the professional sphere**

8 weeks | 8-10 hours a week | Online

Key Takeaways

This course helps participants become leaders in the grand strategic shift that faces organizations in virtually every industry.

At the end of this course, you will develop the following skills:

01 **Familiarize yourself** with intelligent transactions and contracts via Ethereum.

02 **Understand the architecture of the cloud** and deep knowledge of GitHub in order to gain the maximum benefits of your platforms and tools.

03 **Put into practice the acquired knowledge** to create a WebSocket chat application.

04 **Participate in the debate** regarding potential uses of the Internet of Things (IoT) applied to the business field.

05 **Discover of the long transformative legacy** that artificial intelligence is leaving on industry and education.

06 **Analyze access to your private data** during web navigation with the knowledge of detection and defense against the main threats against cybersecurity.



Participant Profile

This course is directed toward those who want to become certified professionals in the grand global phenomenon of digital transformation in business. The careers which will benefit most from the knowledge and skills gained in this course are:

Technical professionals who want to understand and participate in digital transformation processes in their departments.

CEOs, managers, and other directors who lead organizations and teams whose responsibilities lie where business and technology intersect.

Other technical profiles who are interested in the fundamentals and the potential of these new technologies.

All professionals who want to spearhead new projects and introduce further innovations within their organizations.

There are no prerequisites to complete this course, however it is recommended to have a general understanding of the digital technologies entailed.

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Instructors



Dr. Abel Sanchez

Executive Director of MIT's Geospatial Data Center, Research Scientist; Center for Complex Engineering Systems, Sociotechnical Systems Research Center, under the Schwarzman School of Computer Science.

Dr. Sanchez holds a Ph.D. from the Massachusetts Institute of Technology (MIT). He is the Executive Director of MIT's Geospatial Data Center (GDC). His areas of specialty include the Internet of Things (IOT), Big Data, Cybersecurity, and Digital Innovation. He teaches graduate courses in Data Science, Cybersecurity, and Innovation. For the past eight years his research has focused on architecting large-scale computation.

In IOT, Dr. Sanchez led the global network architecture for the Internet of Things at MIT. The design addresses large-scale computation. Compared to the largest numbers in the world, 2 billion computers, 7 billion phones, and 7 billion people, IOT is orders of magnitude bigger. In similar work, Dr. Sanchez directed simulation of the U.S. critical infrastructure with the National Infrastructure Simulation and Analysis Center (NISAC).ia.

In data science, early work included supply chain information engineering, analytics, simulation, and visualization with Wal-Mart, Kraft, and SAP. Dr. Sanchez extended his work to global anti-counterfeiting efforts with Johnson & Johnson, SAP, and Altria. Helping Altria scale track-and-trace using the Electronic Product Code Information Services standard from RFID. The work produced a global infrastructure used by industry and government to this day.

In enterprise computing, Dr. Sanchez led the design of a global data infrastructure simulator, modeling follow-the-sun engineering, to evaluate the impact of competing architectures on the performance, availability and reliability of the system for Ford Motor Company. The simulator modeled user actions, applications, background processes, network load, servers, storage, and global data centers. The work identified data center reductions opportunities estimated at a billion dollars in savings.

In cyber security, Dr. Sanchez directed impact analysis of large-scale cyber-attacks designing Cyber Ranges for the Department of Defense (DOD). Conducting repeatable experiments in impact analysis and the ability to model the cyber environment in a highly portable fashion. Looking at the insider threat Dr. Sanchez led the DOD security study on Enterprise Resource Planning systems across the United States Armed Forces.

In password security, Dr. Sanchez led the design of a password firewall (negative authentication) for the Intelligence Advanced Research Projects Activity (IARPA) agency. The Negative Filtering or Negative Authentication (NA) approach utilizes a form of complement profiles which resembles the censoring and maturation process of T- cells in the immune system.

In machine learning, addressing financial fraud, Dr. Sanchez designed a situational awareness framework that exploits different perspectives of the same financial data and assigns risk scores to entities (e.g. payment documents) to improve false positive ratios and assist the identification of fraudulent activity in huge and unlabeled financial data in collaboration with Accenture.

In physical security, Dr. Sanchez is developing algorithms to assess risk in the integration of information technology (IT) and operations technology (OT).
Dr. Sanchez is the founder and Chief Software Architect of the Open Source RFID platform project. Dr. Sanchez' software systems are used by Samsung, NEC, NTT, Hitachi, Motorola, SAP, IBM, and Microsoft. Other software initiatives are in use by Sandia National Laboratories, MIT, and by several organizations in East Asia and Europe.
Dr. Sanchez advises companies in the Americas, Europe, Middle East, and Asia.

> "We cannot bet on the artificial intelligence, or the human being separate. We must get the fusion of both."

Affiliations

- MIT Geospatial Data Center (GDC)
- MIT Sociotechnical Systems Research Center (SSRC)
- MIT Institute for Data, Systems, and Society (IDSS)
- MIT Center for Complex Engineering Systems (CCES)
- MIT Center for Computational Science and Engineering (CCSE)
- MIT Consortium for Improving Critical Infrastructure Cybersecurity (IC3)
- MIT Schwarzman School of Computer Science
- MIT AutoID Laboratory

Areas of Interest and Expertise

- Machine Learning
- Cyber/Physical Security
- Enterprise Computing
- Data Science
- IOT

Courses Taught with MIT Professional Education

- Applied Generative AI for Digital Transformation
- Blockchain: Disruptive Technology
- Cloud & DevOps: Continuous Transformation
- Data Leadership: Transforming the Corporation´s Operations, Management, and Mindset to Leverage Data, AI, and Cloud Computing
- Digital Transformation: From AI and IOT to Cloud, Blockchain, and Cybersecurity



Prof. John R. Williams

Professor of Information Engineering, Civil and Environmental Engineering and Director of MIT Geospatial Data Center, and a faculty member in the Center for Computational Science and Engineering part of the Schwarzman School of Computer Science.

John Williams holds a BA in Physics from Oxford University, a MS in Physics from UCLA, and a Ph.D. in Numerical Methods from University of Wales, Swansea. His research focuses on the application of Exascale computation to problems in cyber-physical systems, security and energy. His work on fault-tolerant computing using container migration won the IEEE, High Performance Extreme Computing Conference award for Best Innovation in 2019. He is director of MIT's Geospatial Data Center and from 2006-2012, was Director of the MIT Auto-ID Laboratory, where the Internet of Things was invented.

He is author or co-author of over 250 journal and conference papers, as well as the books on Rock Mechanics and RFID Technology.

He contributed to the 2013 report for the UK Office for Science Foresight Project- The Future of Manufacturing. Alongside Bill Gates and Larry Ellison, he was named as one of the 50 most powerful people in Computer Networks. He consults to companies including Accenture, Schlumberger, Shell, Total, Exxon, SAP Research, Microsoft Research, Kajima Corp, US Lincoln Laboratory, Sandia National Laboratories, US Intelligence Advanced Research Projects Activity, Motorola, Phillip-Morris Inc., Ford Motor Company, Exxon-Mobil, Shell, Total, and ARAMCO.

His international collaborations include Oxford and Cambridge Universities, HKUST, KACST, Alfaisal University, PolyU Hong Kong, Imperial College of Science and Technology UK, Malaysia University of Science and Technology (MUST), and Masdar Institute of Science and Technology Abu Dhabi. He organized the first Cyber-Physical Security Conference in the UK (2011) and along with Dr. Sanchez, he runs the MIT Applied Cyber Security Professional Education summer course. At MIT he teaches courses Architecting Software Systems (MIT 1.125) and Engineering Computation and Data Science (MIT 1.00/1.001).

In data engineering and data science, early work included simulation of Ford's global network, analysis of SAP smart grid billing system. For Altria, he analyzed the performance of item level tagging and also their implementation of an anti-counterfeiting system using the Electronic Product Code (EPC).

In password security, Dr. Williams was PI that developed the algorithms for a negative password authentication system for the Intelligence Advanced Research Projects Activity (IARPA) agency.

Dr. Williams advises companies in the Americas, Europe, Middle East, and Asia.

> "Once you climb that first mountain of knowledge, you will be able to teach yourself many things."

Affiliations

- MIT Department of Civil and Environmental Engineering
- MIT Center for Computational Science and Engineering (CCSE)
- MIT Schwarzman School of Computer Science
- MIT Geospatial Data Center (GDC)
- MIT Auto-ID Laboratory
- MIT Center for Complex Engineering Systems (CCES)
- MIT Consortium for Improving Critical Infrastructure Cybersecurity (IC3)
- MIT System Design and Management Program

Areas of Interest and Expertise

- Information Technology,
- Cyber/Physical Security
- Web-Based Education Technology
- Large Scale Network Simulation
- GeoNumerics of Granular and Powder Systems
- Modern Software Architecting and Cyber Security
- Web Services and Distributed Computing
- Discrete Element Simulation and Analysis of Discontinua

Courses Taught with MIT Professional Education

- Applied Generative AI for Digital Transformation
- Blockchain: Disruptive Technology
- Cloud & DevOps: Continuous Transformation
- Data Leadership: Transforming the Corporation´s Operations, Management, and Mindset to Leverage Data, AI, and Cloud Computing
- Digital Transformation: From AI and IOT to Cloud, Blockchain, and Cybersecurity

Course Outline

Module 1
Blockchain: Introduction and Applications

1. Introduction to Blockchain
2. The Evolution of the Internet and the Emergence of Blockchain
3. Key Concepts of Blockchain
4. Smart Contracts: Introduction and Evolution
5. Blockchain and the Challenge of Poverty
6. Identity
7. Initial Coin Offering (ICO)
8. Decentralized Autonomous Organizations
9. Autonomous Cities
10. Blockchain Applications: Architecture
11. Digital Cats and Analog Cows
12. Conclusions

Module 2
Key Concepts of Blockchain

1. History of Money and Transactions
2. Technological Context and Drivers of Change
3. Challenges: System Coordination and Security
4. The Architecture of Blockchain
5. Blockchain Operations
6. Smart Contracts
7. New Horizons

Module 3
Cloud

1. Introduction to Cloud
2. Introduction to Cloud Services
3. Everything as a Service (XaaS)
4. Programmable Virtual Machines
5. Containers
6. Microservices
7. Serverless Models

Module 4
Artificial Intelligence

1. Introduction to Artificial Intelligence
2. Artificial Intelligence
3. Language Translation
4. Machine Evolution
5. Disruptions in the Business World
6. Augmented Reality
7. Managing Data with Artificial Intelligence and Machine Learning
8. Machine Learning
9. Building Capabilities: Data and the Data Science Team

Module 5
The Internet of Things

1. Introduction to the Internet of Things (IoT)
2. Radio Frequency Identification (RFID)
3. Facebook, Amazon, Apple, Netflix, Alphabet, and Google (FAANG)
4. Cybersecurity
5. Web 4.0
6. Protocols
7. The Attack on Ukraine's Power Grid and the IoT
8. Conclusions

Module 6
Cybesecurity

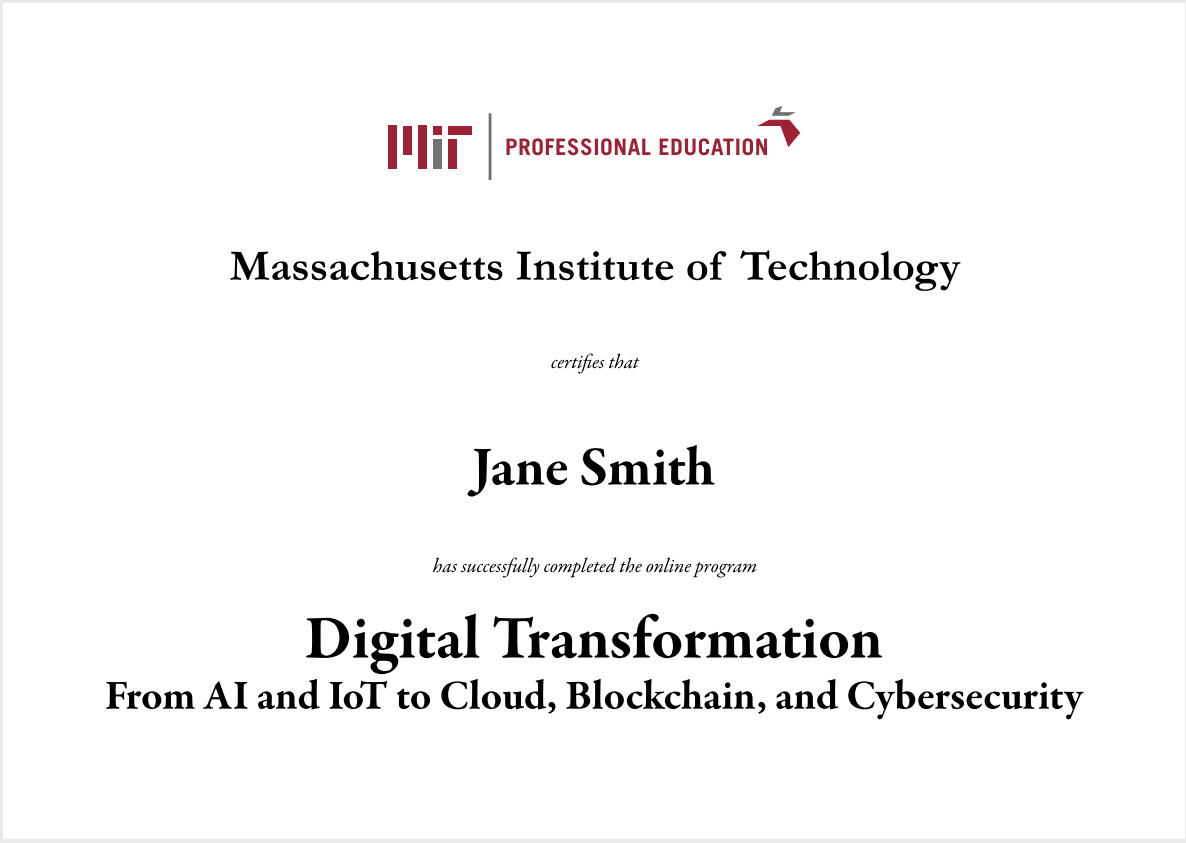
1. Cybersecurity Today
2. The Professionalization of Cyberattacks
3. Passwords and Cybersecurity
4. Browser Privacy
5. Visualize the Threat
6. IoT and Cybersecurity
7. Common Security Failures
8. Finding People Trained in Cybersecurity
9. Some Final Comments About Cybersecurity

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Certificate of Completion

All participants who successfully complete **Digital Transformation: From AI and IoT to Cloud, Blockchain, and Cybersecurity** will receive an MIT Professional Education Certificate of Completion.

In order to obtain CEUs, participants must complete a required CEU accreditation form. CEUs are calculated based on the number of learning hours in each course.



*The Continuing Education Unit (CEU) is defined as 10 contact hours of ongoing learning to indicate the amount of time they have devoted to a non-credit/non-degree professional development program. To understand whether or not these CEUs may be applied toward professional certification, licensing requirements, or other required training or continuing education hours, please consult your training department or licensing authority directly.



Welcome to the gateway

to MIT knowledge and expertise for professionals around the world.

155

Participants from 155 countries

+1600

More than 1,600 online activities

92%

92% of Professional Alumni rated their experience as outstanding

About MIT Professional Education

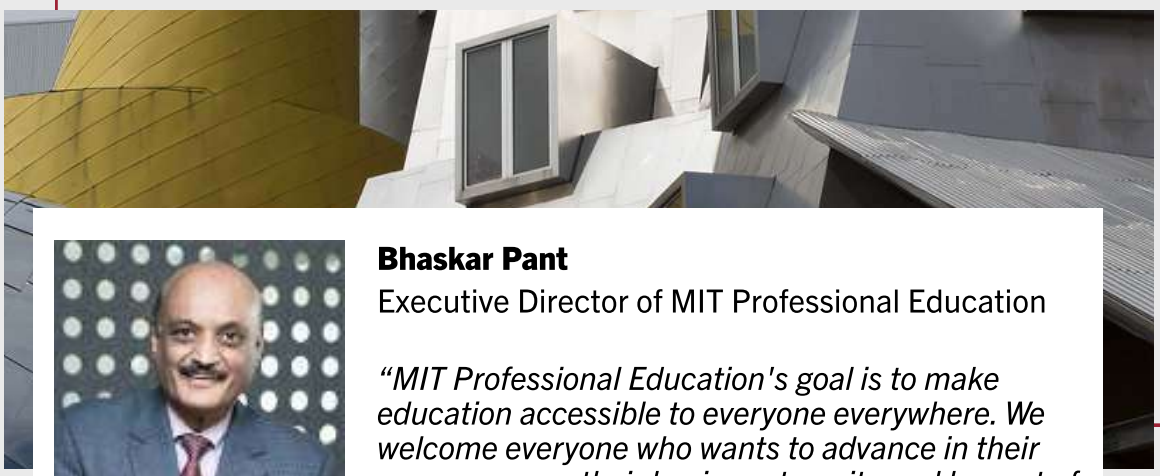
MIT Professional Education fosters the development of innovative leaders equipped to address complex problems globally

MIT Professional Education provides continuing education courses and lifelong learning opportunities for science, engineering, and technology professionals at all levels, from around the world. Worthy of note is that **MIT professors and lecturers lead and teach all MIT Professional Education offerings**. Some of our programs have a long history. Others are relatively new.

Our Digital Plus Programs go beyond online, blending innovative content delivered using the best of online technology and traditional classroom instruction in a flexible, collaborative learning environment.

MIT Professional Education is central to MIT's vision. It fulfills the mandate to connect practitioner-oriented education with industry and incorporate industry feedback and knowledge into MIT education and research.

The Institute is committed to generating, disseminating, and preserving knowledge while working with others to apply this **knowledge for the benefit of humankind.**





Bhaskar Pant
Executive Director of MIT Professional Education

"MIT Professional Education's goal is to make education accessible to everyone everywhere. We welcome everyone who wants to advance in their career or grow their business to unite and be part of the MIT Professional Education all-inclusive community."

Corporate Courses

At MIT Professional Education, we are committed to accompanying organizations as they aspire to provide their professionals with the ongoing education critical for **success in today's competitive environment**. That's why we offer plans for companies who wish to train their professionals with us.

Which program best suits your organization's needs?

1

Standard Program:
Teams or groups of at least 10 people will obtain special conditions in their registration for any of our programs.

2

Custom Programs:
Custom-made to meet the specific needs of each organization and its professionals.

3

Private Cohorts:
Available in Open Enrollment courses with a minimum of participants.

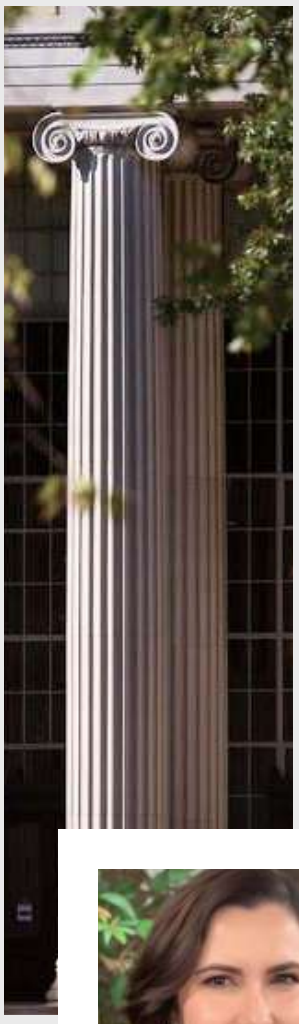


For more information on our Programs for Organizations, please contact us at: **professionalprograms@mit.edu**

The Beyond Online Methodology

MIT Professional Education is **revolutionizing the digital learning experience**. In fact, we are no longer talking about online-only, but rather an **interactive and collaborative learning experience** that's digitally powered: a **Beyond Online experience.**

You will have access to one of the most innovative e-learning platforms that utilizes the latest technology.You will also have an opportunity to meet and work with some of the world's leading subject matter experts in their respective fields.



With MIT Professional Education, you'll have an opportunity to:

- Attend select live webinars with MIT Instructors.
- Collaborate and engage with world-renowned MIT Instructors.
- Learn from expert learning facilitators, guest speakers, and subject matter experts from industries.
- Weekly live webinars from learning facilitators.
- Fulfill your reskilling and upskilling needs in this fast-changing environment.
- Master the technical and human skills necessary for effective leadership.
- Meet and network with other program participants from all around the world.



Clara Piloto
Director of Global Programs at MIT Professional Education

"We are currently amid Digital Transformation and Industry 4.0. It is now necessary to understand AI, IoT, cloud, blockchain, and cybersecurity to maintain a competitive edge professionally. Leverage the digital transformation to spark innovation and growth in your organization and in your profession."

Benefits of Joining the Alumni Community



MIT Professional Education offers **a number of benefits for participants** who successfully complete our courses:

1. **15% discount** for MIT Professional Education online courses and short-duration, in-person programs.
2. **Continued access to the virtual campus for six months after completion** of the course.
3. **Exclusive announcements** of new courses, programs, and events through our **virtual Alumni Campus: Professional Network.**
4. Invitation to the **MIT Professional Education Alumni LinkedIn group.**
5. **Networking opportunities** with other MIT Professional Education Alumni.

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PROFESSIONAL EDUCATION
Digital Plus+ Programs

Are you ready to take charge of the latest technologies?

Frequently Asked Questions

global alumni
-excellence or nothing-

For more information, please contact us:
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Email: **professionalprograms@mit.edu**
Website: **https://professionalprograms.mit.edu**